Manual

HYDRA Interfacing Module to SAP CO-ILV

SAP-COILV 8.2

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Contents

1 HYDRA Interfacing Module to SAP CO-ILV

Summary

Use options

The SAP-COILV interface allows to perform SAP CO uploads via the SAP CO interface. The upload can be made for both, the direct and indirect activity allocation.

This interface enables the user to directly debit orders and primarily CO internal orders and cost centers and to compensate between cost centers and/or to transfer the activities performed by the cost centers to SAP to compensate and to assign them to accounts. This means that it is possible to transfer the data collected in MES directly to the CO.

Implementation notes

Use the function package if

- you wish to transfer CO internal orders directly from the SAP CO to the MES;
- you wish to collect MES confirmations of a CO internal order and to confirm them to SAP CO;
- you wish to enter the MES times referred to the individual employees and/or cost centers and then to compensate them directly or indirectly in SAP CO.

Integration

The function package uses the BDE data.

Scope of functions

- Confirmation of direct activity allocations
    - Confirmation of direct activity allocation via BAPI AcctngActivityAlloc.Post
    - Confirmation of indirect activity allocation via BAPI AcctngSenderActivity.Post

2 Mapping CO-ILV in HYDRA

In the course of a connection of HYDRA to SAP CO, HYDRA must collect CO-relevant data and upload them to SAP. As data basis serve both, the CO internal orders transferred from SAP to HYDRA and orders that were created in HYDRA and that include the specific data required for the confirmation in accordance with the creation convention.

The trigger to download CO internal orders comes from R/3. The data are transferred in an IDoc (intermediate document) and maintained in HYDRA. In principle, internal orders are transferred by two methods. On the one hand, it is possible to use the default interface HR-PDC to transfer internal orders to HYDRA. It is true that this method uses the SAP default but in the same time it has the disadvantage that the used structure is very narrow and does not include several often required data types (such as start and end date or scheduled workplace).

Next to this method, it is also possible to transfer CO-internal orders in a customer-specific IDoc in the HYDRA BAPI format. To use this method, a customer-specific function module is necessary which selects the required data in SAP, transfers them to an IDoc and passes them then on to HYDRA.

To realize confirmations referred to cost centers to SAP CO there is also the possibility to create overhead cost orders in HYDRA and to realize the confirmation to specific cost centers. This is only possible if the sending and/or receiving cost center and an activity type are stored to HYDRA.

The upload of the confirmations is controlled via HYDRA in accordance with the requirements specified by the user. In these instances is it not important whether these are confirmations of CO-internal orders or for orders created in HYDRA.

To realize the communication with the BDE subsystems, SAP provides several standard-BAPIs/ IDocs via the CO-interface. The following BAPIs/ IDocs are used:

Download of CO-internal orders (customer-specific):

IDoc type:	ZHYDRA_CO_ORDER
Message type:	ZHYDRA_CO_ORDER
Segment type:	Z1BAPI000

Upload of confirmations (direct activity allocation):

IDoc type:	ACC_ACT_ALLOC02
Message type:	ACC_ACT_ALLOC
Segment type:	E1ACC_ACT_ALLOC000 E1BPDOCHDRP000 E1BPAAITM002

Upload of confirmations (indirect activity allocation):

IDoc type:	ACC_SENDER_ACTIVITIES01
Message type:	ACC_SENDER_ACTIVITIES
Segment type:	E1ACC_SENDER_ACTIVITIES E1BPDOCHDRP000 E1BPIAITM000

3 Data container Z2BAPI000

The base structure Z2BAPI000 is used as basis to transfer data from external systems. Depending on the data type, different posting and IDoc types will be used. The segment structure Z2BAPI000, however, is always the same.

To create the segment name Z2BAPI000 in SAP, it must also be generated in SAP according to the scheme Z1 <Segment name>. Versioning in SAP outbound processing is then used to generate the segment names of the form Z2 <Segment name><Version>.

Example: **Z1**BAPI becomes **Z2**BAPI**000**

The segment consists of the following 3 fields:

Field	Type	Length	Description
TRANSACTION	CHAR	20	Name of the transaction in HYDRA (Dialog identification in HYDRA)
DESC	CHAR	40	Comment
DATA	CHAR	940	Dialog data string for HYDRA

The TRANSACTION field contains the control command that is also transferred in the DATA field. It has no function here and is only used for information purposes.

The DESC field can be used to transfer a comment text describing the operation.

The DATA field is used to transfer the user data. The transfer is realized in the HYDRA HYBAPI format, i.e. a dialog data string composed of the control command and the user data is transferred. The control command is always transferred with the "DLG=" acronym followed by the command itself. The control command is followed by several data identified by a dialog identification and separated by "|". The dialog string itself must be terminated by a pipe "|".

4 Structure of the DLG Format

Basics of HYDRA BAPI

Data is always posted to the database in accordance with basic guidelines ensuring their consistency and uniformity. This is why any writing access to the database is performed by programs providing a uniform interface to this end.

This means that all writing accesses to the HYDRA database irrespective of whether these are called via HYDRA applications or external applications/ systems are executed by a program with a defined interface.

This is mainly the HYDRA BAPI. It is used in the course of the master data transfer in order to transfer and post data provided by and processed in external systems to HYDRA.

BAPIs and dialog commands

Essentially, there is for each object (that can be maintained using the MOC) such a BAPI in HYDRA. Objects in this sense may be (production) orders or master data records. There are always different methods to access such an object. In the easiest case this is a method to create (INSERT), modify (UPDATE) or delete data records (DELETE).

In more complex cases and/or when this is requested by the application, also different methods are implemented. This may be modifying methods comprising an insertion or modification or additional application-specific methods.

Such a BAPI is called by a so called dialog command. This command is comprised of:

<Object>**.**<Method>

This is an exemplary (and incomplete) overview of the available objects and their selected methods

Object	**Methods**	**Comment**
ANR	INSERT UPDATE DELETE MODIFY	The ANR object designates the order.

Object	Methods	Comment
MNR	INSERT UPDATE DELETE	The MNR object designates machines/ workplaces.
FERTVAR	INSERT UPDATE DELETE	The FERTVAR object designates production variants.
RES	INSERT UPDATE DELETE	The RES object designates the resources of the module WRM and DNC.

Dialog data strings

After the initial BAPI call using the command, the use data will be transferred in a so-called dialog string or dialog data string. The use data in a dialog string are clearly identified by indicators, also designated as acronyms.

Such an acronym may represent at least one database field or also have controlling effects on postings. The acronym is always followed by the equal sign "=" and the value transferred for this acronym. The individual acronyms and their values are separated by pipes "|" from each other and from the dialog command.

Example:

```
DLG=FERTVAR.INSERT|FERTVAR.ATK=BLOO01052225000O00|FERTVAR.MGRP=BW2000|
FERTVAR.RESTYP=WNR|FERTVAR.RES=BLOO01052225000O00 2|
FERTVAR.SZY=17143|FERTVAR.TLG=2|
FERTVAR.BEM=BLOO01052225000O00\|rose\|2\|rose\|BW2000|
FERTVAR.VER=1|FERTVAR.STA=F|FERTVAR.FIR:ATK=0|
```

Data formats/ mandatory acronyms

The descriptions of the acronyms are based on the following data types:

Type	Description
CHAR x	For the data type CHAR the information will be aligned to the left; unnecessary positions will be filled with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. For the NUMC data type only digits are allowed (ASCII-digits 30 hex to 39 hex). The numbers will be aligned to the right and unnecessary positions will be filled with zeros. Example: "00000002"
DEC x.y	Numeric field of the length x contains y decimal places. A data field in the HYDRA format is **preceded** by a sign ("+" or "-") and it contains a decimal point. Empty places must be filled with zeros. e.g. DEC 13.3: **-**1234567890.123

Each BAPI call ***must*** contain the following header data in the dialog data

<table>
<tr><th>Identification</th><th>Content</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>{BAPI call}</td><td colspan="2">Dialog identification: This dialog identification indicates the desired BAPI call</td></tr>
<tr><td rowspan="5">USR</td><td rowspan="5">NUM 4</td><td colspan="2">HYDRA user: This Hydra user number uniquely identifies a HYDRA client:</td></tr>
<tr><td>MOC:</td><td>USR = 20000 + MOC number
USR = 20000 + MOC</td></tr>
<tr><td>LAN terminal (LANT)
FB terminal (FBT):</td><td>USR = 2000 + terminal number
USR = 2000 + TNR</td></tr>
<tr><td>External terminals</td><td>USR = 3000 ... 3999</td></tr>
<tr><td>MLE-MDM</td><td>USR=9999</td></tr>
<tr><td>DAT</td><td>{mm/dd/yyyy}</td><td colspan="2">Date: current date in the format mm/dd/yyyy
"Today" can be used as placeholder for the dynamic determination.</td></tr>
<tr><td>ZEI</td><td>{seconds}</td><td colspan="2">Time: current time in the seconds format
"Now" can be used as placeholder for the dynamic determination.</td></tr>
</table>

Depending on the BAPI call, additional identifications must/ may be entered.

Data objects with files

Only the file names will be indicated in the dialog string for such objects that contain files in addition to the data fields of dialog data strings, e.g. document resources or DNC resources. The files themselves will be stored to defined data areas. The data import consists of two steps: Dialog data strings and files.

Dialog data strings - acronyms

The acronym to indicate the file is a field of the field type CHAR 128 that includes the file name. In most of the cases the name is only indicated without path - please see the documentation for the BAPI concerned.

Example: RES.SPEICHORT:DATA includes the file name without path and without extension of the attached DNC file. The storage location and the extension are defined before in the system via the resource type.

File format:

The file format is not important for the storage in HYDRA. The file will be stored to the specified storage location. The application will then interpret this file. For the import of master data it must be taken into account that the file must be stored to the directory specified for the application.

Example DNC files: The DNC type defines in which folder the files are and how they must be stored and interpreted.

Multilingual database contents

As part of SIS-HLM, there is now the possibility to define descriptive texts in several languages for specific objects in the database. Provided that this function is enabled on the system, these columns may generally also be filled by using the master data import. Please note the following:

- Specify the target language
  The target language can be transferred as additional acronym in the dialog data strings.
  Example:
  Machine master data is to be transferred. English (EN) is defined with language index 2 in the system. The dialog data string to transfer this data has to be structured as follows:

  DLG=MNR.INSERT|…|MNR.MNR=<Machine>|MNR.BEZK=English description|…|LANG=2|…

- Only one language can be transferred every time an import is started.

This means, that two or more import runs might be required, subject to the number of configured languages. Please note the following:

- The first import has to be performed using the *.INSERT method.
  This rule can be ignored if there is a method "*.MODIFY" for the object. As in this case, the system decides whether an INSERT or an UPDATE is to be performed.
- All other imports need to be performed by way of the method "*.UPDATE" indicating all key fields pertaining to the object, the language-dependent description and the target language using the acronym "LANG=n".
- If the system uses a separately generated, internal key for an object, this one has to be determined after the initial creation. This internal key then needs to be provided for the updates that follow.

5 1 Example: Download of CO Internal Order

IDoc ZHYDRA_CO_ORDER

IDoc ZHYDRA_CO_ORDER is used to transfer CO internal orders. It has a simple structure since the user data themselves are transferred to HYDRA in a dynamic dialog string.

Message type:	ZHYDRA_CO_ORDER
IDoc type:	ZHYDRA_CO_ORDER
Segments:	Z2BAPI000

Dialog data string

In HYDRA, one order header is created per CO internal order. Several operations may be allocated to this order header. Order header and operations can be transmitted in the same IDoc if the order header is transferred as first segment.

A dialog data string is composed of the control command and the user data The control command is always transferred with the “DLG=” acronym followed by the command itself.

NOTE:

For all alphanumeric fields, HYDRA does not support specific special indicators. Such as: "%", "\", "/", "|" since they cannot be entered into the collection terminals and will not be supported there.
The signs ";", " “ ", and " ’ " must not be used since they are often interpreted as comment or separation signs and will thus lead to unwanted effects.

Order header

The following commands can be used to transfer CO internal orders to the order header:

- ANR.INSERT → Creation of an order header
- ANR.UPDATE → Modification of an order header (delta download)
- ANR.DELETE → Deletion of an order header (deletion download)

The command "AUNR.MODIFY" is a special one: It checks whether the transferred order number exists already in the order header. If this is the case, the existing data record will be modified (update), otherwise it will be inserted.

- ANR.MODIFY → Creation/ modification of an order header (delta download)

The different commands ensure that the download variants such as delta or deletion download of the PP-PDC interface can be realized.

After the control command, the user data will be transferred. They are presented by the identification (Acronym column) and separated from each other by "|".

What	Acronym	SAP / Value
Order number	AUNR.SAPAUNR	SAP order number
Order type	AUNR.AART	"1" or "4"
PPS Indicator	PPS	"J"
Indicator order header	ANR.ATYP	"AU"

In this example the dialog data string for a delta download will be structured as follows:

```
DLG=AUNR.MODIFY|AUNR.SAPAUNR=<SAP-Auftragsnummer>|AUNR.AART=1|PPS=J
```

Order sequencing

The following commands can be used to transfer CO internal orders to the HYDRA operation structure:

- ANR.INSERT → Creation of an operation
- ANR.UPDATE → Modification of an operation (delta download)
- ANR.DELETE → Deletion of an operation (deletion download)

The command "ANR.MODIFY" is a special one: It checks whether an operation exists already for the transferred order and an operation number. If this is the case, the existing data record will be modified (update), otherwise it will be inserted.

- ANR.MODIFY → Creation/ modification of an operation (delta download)

The different commands ensure that the download variants such as delta or deletion download of the PP-PDC interface can be realized.

After the control command, the user data will be transferred. They are presented by the identification (Acronym column) and separated from each other by "|".

What	Acronym	SAP	Mandatory field
Order number	ANR.SAPAUNR	SAP order number	X
Sequence	ANR.SAPAFOLG	SAP sequence	X
Operation	ANR.SAPVGNR	SAP operation number If not available in this form, it is also possible to transfer "0010".	X
Sub-operation	ANR.SAPUVGNR	SAP sub-operation (if necessary)	X
Plant	ANR.WERK:S	SAP plant	
Workplace	ANR.MNR	SAP work center	X
OP designation	ANR.AGBEZ	e.g. order short text	
Order type	ANR.AART	"1"	X
PPS Indicator	PPS	"J"	X
OP indicator	ANR:ATYP	"OP"	X
Start date	ANR.DATB	ATTENTION: The date must be transmitted in American format: MM/DD/YYYY	
Start time	ANR.ZEIB	Time in seconds	
End date	ANR.DATE	ATTENTION: The date must be transmitted in American format: MM/DD/YYYY	
End time	ANR.ZEIE	Time in seconds	
The indicator can be logged on several times	ANR.OPT:MULTIMNR	"J"	

In this example the dialog data string for a delta download will be structured as follows:

```
DLG=ANR.MODIFY|ANR.SAPAUNR=<SAP order number>|ANR.SAPVGNR=<SAP
transaction number>|ANR.WERKS=<SAP plant>|ANR.MNR=<Workplace>|
ANR.AGBEZ=<OP name>|ANR.AART=1|PPS=J|...
```

6 Example: Using MES-Internal Orders

Usage

HYDRA overhead cost orders are orders that are created in HYDRA. They are not known in SAP. These orders can be used then, for example, if a special service has been rendered by one cost center and will be invoiced to another.

The HKMCO-ILV interface allows to map these processes and using the confirmation, to charge the rendered service directly to the receiving cost center. These are two quick steps to realize this.

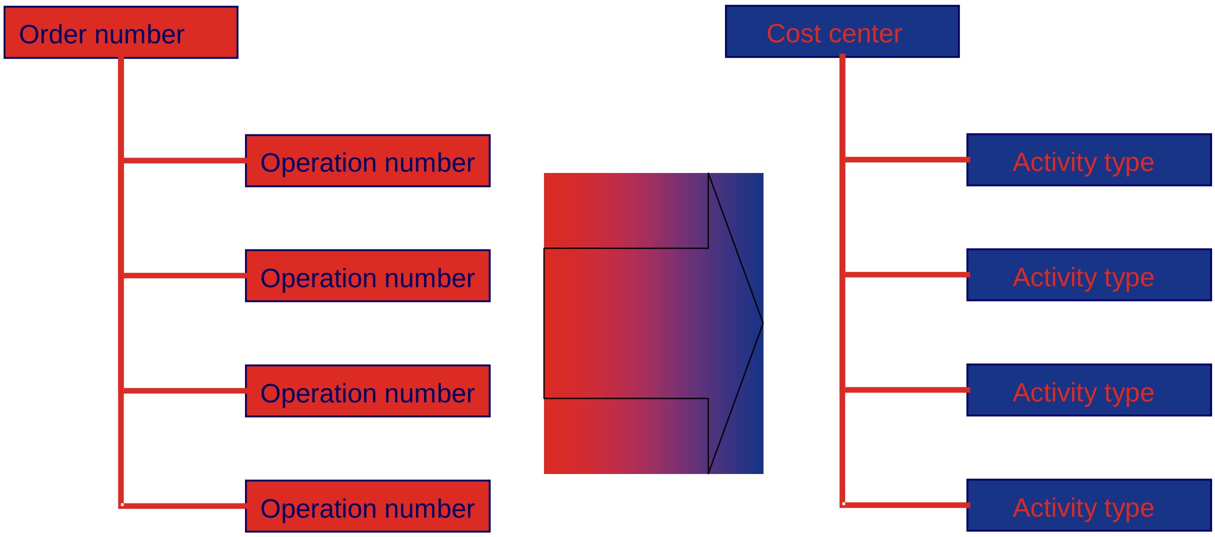
Order number = receiving cost center

In this approach the receiving cost center is part of the order number. The confirmation configuration will define in this case which interval of the order number represents the cost center number.

The activity type is stored to the operation number. Here too, the confirmation configuration can be used to define which interval of the operation number will represent the activity type.

The sending (performing) cost center can either be taken from the workplace or the personnel. To this end, the these data must be maintained in HYDRA.

Based on these conventions, this will lead to the following order structure:

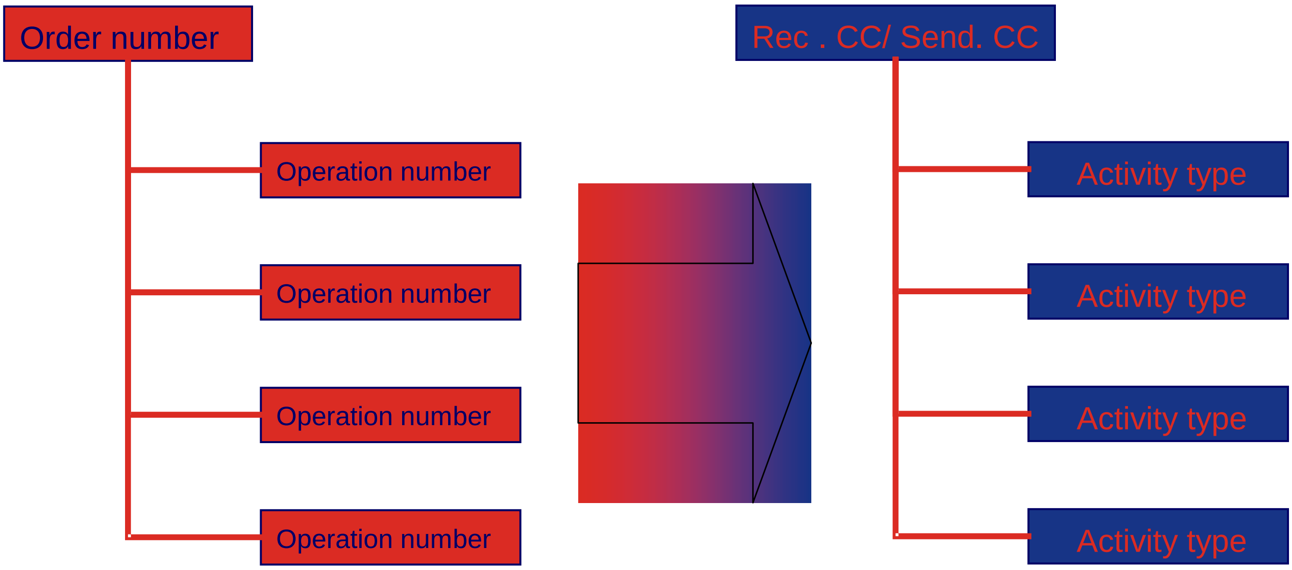


In the realization, special attention must be paid to the number ranges of the other order types known in HYDRA and to the lengths of the order numbers configured in HYDRA.

Order number = sending/ receiving cost center

In another form of mapping the orders can be stored to both - the sending and receiving cost center in the HYDRA order number. Here too, the configuration of the confirmations will define which interval of the order number will represent which cost center.

As in the previous example, the activity type will be stored to the operation number. Based on these conventions, this will lead to the following order structure:



In the realization, special attention must be paid to the number ranges of the other order types known in HYDRA and to the lengths of the order numbers configured in HYDRA.

7 Upload of Confirmations

Summary

Confirmation general

HYDRA BDE confirms to SAP on the basis of time tickets. Transferred are the labor data durations from the B-records (personnel postings). They are transferred to SAP together with the personnel number.

Which cost center (e.g. for staff or the workplace) is transmitted to SAP, whether the receiving object is a CO-internal order or a cost center can be configured in the interface. The configuration is explained in chapter *0 Configuration of Uploads to SAP CO*.

Confirmation - direct activity allocation

Segment E1ACC_ACT_ALLOC000

Field name	T	L	Meaning	Meaning in HYDRA		
IGNORE_WARNINGS	Char	1	Ignore warnings	Not occupied		

Segment E1BPDOCHDRP000

Field name	T	L	Meaning	Meaning in HYDRA		
CO_AREA	Char	4	Controlling area	According to configuration		
DOCDATE	Char	8	Document date	Log. Date		
POSTGDATE	Char	8	Posting date	Set according to shift date of the B-records		
VERSION	Char	3	Version	According to configuration*)		
DOC_NO	Char	10	Document number	Not occupied		
VARIANT	Char	5	Fast document entry of CO-actual postings: Variant	According to configuration*)		
DOC_HDR_TX	Char	50	Document header text	Not used		
USERNAME	Char	12	Name of the user	According to configuration*)		
OBJ_KEY	Char	20	Reference key	Not used		
OBJ_TYPE	Char	5	Reference operation	Not used		
OBJ_SYS	Char	10	Logical system of the original document	Not used		

Segment E1BPAAITM002

Field	T	L	Meaning	Meaning in HYDRA		
SEND_CCTR	Char	10	Sending cost center	According to configuration*)		
ACTTYPE	Char	6	Activity type	According to configuration*)		
SENBUSPROC	Char	12	Sending business process	Not used		
ACTVTY_QTY	Char	17	Activities quantity	According to configuration*)		
ACTIVITYUN	Char	3	Activity unit	According to configuration*)		
ACTIVITYUN_ISO	Char	3	ISO-code unit of measurement	Not used		
PRICE	Char	25	Price total in the currency of the transaction	Not used		
CURRENCY	Char	5	Currency key	Not used		
CURRENCY_ISO	Char	3	Iso-code currency	Not used		
POS_OUTQTY	Char	17	Posted output quantity	Not used		
POSTOUTUN	Char	3	Posted output unit	Not used		
POSTOUTUN_ISO	Char	3	ISO-code unit of measurement	Not used		
PERSON_NO	Char	8	Personnel number	Personnel number		
SEG_TEXT	Char	50	Segment text	Not used		
REC_CCTR	Char	10	Receiving cost center	According to configuration*)		
REC_ORDER	Char	12	Receiving order	According to configuration*)		
REC_WBS_EL	Char	24	Receiving project structure scheduling element	Not used		
RECSALEORD	Char	10	Receiving sales order	Not used		
RECITEM	Char	6	Position number in recipient sales order	Not used		
RECCOSTOBJ	Char	12	Receiving cost object	Not used		
RECBUSPROC	Char	12	Receiving business process	Not used		
REC_NETWRK	Char	12	Receiving network	Not used		
RECOPERATN	Char	4	Receiving network operation	Not used		
RECRUNSCHD	Char	12	Receiving repeat order	Not used		
MATERIAL	Char	18	Receiving material	Not used		
PROD_VERSN	Char	4	Production version of the recipient material	Not used		
PLANT	Char	4	Plant of the recipient material	Not used		
RECPRCMTPROC	Char	12	Receiving procurement process	Not used		
ITEMNO_ACC	Char	10	Position number of the accounting document	Not used		
REC_CALC_MOTIVE	Char	2	Recipient calculation motive	Not used		
RECACTTYPE	Char	6	Receiving activity type	Not used		
SRE_COMP_CODE	Char	4	Company code of the sending real estate object	Not used		

Field	T	L	Meaning	Meaning in HYDRA		
SRE_BUS_ENT	Char	8	Sending business unit real estate	Not used		
SRE_PROPERTY	Char	8	Sending lot of land real estate	Not used		
SRE_BUILDING	Char	8	Sending building real estate	Not used		
SRE_RENT_UNIT	Char	8	Sending rental unit real estate	Not used		
SRE_LEASE	Char	13	Sending rental agreement real estate	Not used		
SRE_MGMT_CON	Char	13	Sending administration agreement	Not used		
SRE_INC_EXP	Char	4	Sending incidental costs key real estate	Not used		
SRE_SETT_UNIT	Char	5	Sending accounting unit real estate	Not used		
SRE_REF_DATE	Char	8	Sending reference date settlement real estate	Not used		
SRE_CON_NO	Char	13	Sending contract real estate	Not used		
RRE_COMP_CODE	Char	4	Company code of the receiving real estate object	Not used		
RRE_BUS_ENT	Char	8	Receiving business unit real estate	Not used		
RRE_PROPERTY	Char	8	Receiving lot of land real estate	Not used		
RRE_BUILDING	Char	8	Receiving building real estate	Not used		
RRE_RENT_UNIT	Char	8	Receiving rental unit real estate	Not used		
RRE_LEASE	Char	13	Receiving rental agreement real estate	Not used		
RRE_MGMT_CON	Char	13	Receiving administration contract	Not used		
RRE_INC_EXP	Char	4	Receiving incidental costs key real estate	Not used		
RRE_SETT_UNIT	Char	5	Receiving accounting unit real estate	Not used		
RRE_REF_DATE	Char	8	Receiving reference date settlement real estate	Not used		
RRE_CON_NO	Char	13	Receiving contract real estate	Not used		
MATERIAL_EXTERNAL	Char	40	Long material number (future development) for the field MATER	Not used		
MATERIAL_GUID	Char	32	External GUID (future development) for the field MATERIAL	Not used		
MATERIAL_VERSION	Char	10	Version number (future development) for the field MATERIAL	Not used		

Confirmation - indirect activity allocation

Segment E1ACC_SENDER_ACTIVITIES

Field name	T	L	Meaning	Meaning in HYDRA		
IGNORE_WARNINGS	Char	1	Ignore warnings	Not occupied		

Segment E1BPDOCHDRP000

Field name	T	L	Meaning	Meaning in HYDRA		
CO_AREA	Char	4	Controlling area	According to configuration*)		
DOCDATE	Char	8	Document date	Log. Date		
POSTGDATE	Char	8	Posting date	Set according to shift date of the posting records		
VERSION	Char	3	Version	according to configuration*)		
DOC_NO	Char	10	Document number	Not occupied		
VARIANT	Char	5	Fast document entry of CO-actual postings: Variant	According to configuration*)		
DOC_HDR_TX	Char	50	Document header text	Not used		
USERNAME	Char	12	Name of the user	According to configuration*)		
OBJ_KEY	Char	20	Reference key	Not used		
OBJ_TYPE	Char	5	Reference operation	Not used		
OBJ_SYS	Char	10	Logical system of the original document	Not used		

Segment E1BPIAITM000

Field name	T	L	Meaning	Meaning in HYDRA		
SEND_CCTR	Char	10	Sending cost center	according to configuration*)		
ACTTYPE	Char	6	Activity type	according to configuration*)		
SENBUSPROC	Char	12	Sending business process	Not used		
ACTVTY_QTY	**Quan**	15.3	Activities quantity	according to configuration*)		
ACTIVITYUN	Char	3	Activity unit	according to configuration*)		
ACTIVITYUN_ISO	Char	3	ISO-code unit of measurement	Not used		
POS_OUTQTY	**Quan**	15.3	Posted output quantity	Not used		
POSTOUTUN	Char	3	Posted output unit	Not used		
POSTOUTUN_ISO	Char	3	ISO-code unit of measurement	Not used		
PERSON_NO	Numc	8	Personnel number			
SEG_TEXT	Char	50	Segment text	Not used		

8 Configuration of Uploads

Summary

Configuration of Uploads to SAP CO

SAP CO has been designed as dynamically as possible to be able to meet the different requirements and situations. The table SAP_RCK_DATA_CONF is used for configurations. Parameters, program parameters and field parameters are defined in this table.

Parameters:

Statistical values required for the generation of the IDoc can be defined using these parameters (e.g. the company code).

Program parameters

Program parameters control program processing. These parameters can be used, for example, to define that only orders of a specific HYDRA order type are uploaded/confirmed.

Field parameters:

Field parameters define from which HYDRA data model fields values relevant for the upload are taken. Moreover, for certain field parameters it is possible to define in which length and as of which position these values are to be taken.

All parameters are saved as variant. This means that the upload program is called up with the corresponding message type and the variant and the settings defined in this variant are used for the upload.

Please note: Important: When calling the program MYERPRCK to generate upload records, the variant is to be sent as well so that this variant will be forwarded as parameter to the user exit.

e.g. sh.exe ./myerprck.scr /MESTYP=ACC_ACT_ALLOC /KAT=GK **/UE_PARAMS="VARIANTE=SAP"**

Parameters

Key type	Parameter	Description	Values
P	CO_AREA	The content of this field of the segment E1BPDOCHDRP000 of the upload IDoc is transferred to the field CO_AREA. It includes the SAP controlling area.	e.g. 1000

Key type	Parameter	Description	Values
P	VERSION	The content of this field of the segment E1BPDOCHDRP000 of the upload IDoc is transferred to the field VERSION.	e.g. 000
P	VARIANT	The content of this field of the segment E1BPDOCHDRP000 of the upload IDoc is transferred to the field VARIANT. It includes the specific upload variant in SAP.	e.g. SAP02
P	ACTIVITYUN	Determines the upload unit	SEK second MIN minute M minute STD hour (Default) H hour

Program parameter overview

Key type	Parameter	Description	Values
PP	NOSAP	If this parameter is set only orders will be uploaded that are not known in SAP	-
PP	ONLYSAP	If this parameter is set only orders will be uploaded that are not known in SAP. →**default setting**	-
PP	LOGSYS	The logical system specified here determines the communication user from the configuration tables of the HYDRA mySAP communication. This user is also entered as the user of data records in SAP.	Logical system of the mySAP communication
PP	MAX_SEG	Maximum number of segments summarized for the upload.	Default 100
PP	USERNAME	User name who is specified as generating user in the data record structure. If no user is specified the communication user indicated in LOGSYS of the logical system is determined and used as the user.	

Field parameters

Key type	Parameter	Description	Values
F	SEND_CCTR	Sending cost center	MNR.KST: cost center of the workplace ADEPRO.ANR: HYDRA order indicating the position from / to PNR.KST: The employee’s cost center

Key type	Parameter	Description	Values
F	ACTTYPE	Activity type	ADEPRO.ANR: HYDRA order number indicating the position from / to PNR.INFOTEXT:1 The employee's activity type from the HR master
F	ACTVTY_QTY	Quantity of the activity type	ADEPRO.EGR:PDAUER: Duration of the B record posting
F	REC_CCTR	Receiving cost center	ADEPRO.ANR: HYDRA order number specifying the position from / to
F	REC_ORDER	Receiving order	AGNR.SAPAUNR: SAP order number of the CO internal order
F	PERSON_NO	Personnel number	ADEPRO.PNR: HYDRA personnel number

Structure of the table SAP_RCK_DATA_CONF

The below table includes the configurations for the upload/confirmation to SAP CO. At the moment, the table can only be edited directly in the database.

Field name	T	L	Meaning	Meaning in HYDRA
KEY_TYPE	Char	2	Key type	Parameter type: „P" Parameter „PP" Program parameter „F" Field parameter
KEY	Char	30	Key	Parameter name, e.g. SEND_CCTR, ONLYSAP
SUBKEY	Char	40	Variant	Variant name by which a configuration is saved and the upload program is started (free text).
MESTYP	Char	30	Message type	Name of the SAP message type
POS_VON	Num		Position from	Field position from which the value is to be taken. If nothing is entered the entire field will be taken over.
POS_BIS	Num		Position to	Field position up to which the value is to be taken. If nothing is entered the entire field will be taken over.
LFD_NR	Num		Consecutive number	Not relevant
VERWEIS	Num		Reference	Unique reference (assigned by the database).
BEARB	Char	10	Editor	Not used

Field name	T	L	Meaning	Meaning in HYDRA
BEARB_DATE	Date		Editing date	Not used
BEARB_TIME	Time		Editing time	Not used
ANLAGE_DATE	Date		Creation date	Not used
ANLAGE_ZEIT	Time		Creation time	Not used
PARAM_STR1	Char	20	Parameter 1	Not used
PARAM_STR2	Char	20	Parameter 2	Not used
PARAM_STR3	Char	40	Parameter 3	Not used
PARAM01	Num		Parameter integer	Not used
PARAM02	Num		Parameter integer	Not used
PARAM01_d	Dec	18,6	Parameter decimal	Not used

9 Application-Relevant Settings in HYDRA

Maintenance of the HYDRA distribution model – outbound processing

Use the HYDRA distribution model to maintain entries for HYDRA outbound processing:

Parameter name	**Value**
To upload the time tickets for direct activity allocation	
Message type	ACC_ACT_ALLOC
Description	CO-ILV – Direct activity allocation
IDoc type	ACC_ACT_ALLOC02
Storage duration	10
Log. target system	Created logical system
Segment name 1	E2ACC_ACT_ALLOC000
To upload the time tickets for indirect activity allocation	
Message type	ACC_SENDER_ACTIVITIES
Description	CO-ILV – indirect activity allocation
IDoc type	ACC_SENDER_ACTIVITIES
Storage duration	10
Log. target system	Created logical system
Segment name 1	E2ACC_SENDER_ACTIVITIES

Scheduler maintenance

The following entries must be made for confirmations/uploads of goods movements in the Scheduler:

Parameter name	**Value**
To upload the time tickets for direct activity allocation – confirmation/upload program	

Parameter name	**Value**
Product key	SAP-COILV
License key	SAP-COILV
Command (Windows):	sh.exe ./myerprck.scr /MESTYP=ACC_ACT_ALLOC /KAT=GK /UE_PARAMS="<configured variant>"
Command (Unix):	./myerprck.scr /MESTYP=ACC_ACT_ALLOC /KAT=GK /UE_PARAMS="<configured variant>"
Comment:	Direct activity allocation HYDRA → SAP
Interval	5
To upload the time tickets for direct activity allocation - upload client	
Product key	SAP-COILV
License key	SAP-COILV
Command (Windows):	sh.exe ./hysapupl.scr /UPLSEGNAM=/UPLSEGNAM=E2ACC_ACT_ALLOC000 /SINGLE_IDOC /SUBLEVEL=2
Command (Unix):	./hysapupl.scr /UPLSEGNAM=/UPLSEGNAM=E2ACC_ACT_ALLOC000 /SINGLE_IDOC /SUBLEVEL=2
Comment:	Direct activity allocation HYDRA → SAP
Interval	5
To upload the time tickets for indirect activity allocation - confirmation program	
Product key	SAP-COILV
License key	SAP-COILV
Command (Windows):	sh.exe ./myerprck.scr /MESTYP=ACC_SENDER_ACTIVITIES /KAT=GK /UE_PARAMS="<configured variant>"
Command (Unix):	./myerprck.scr /MESTYP= ACC_SENDER_ACTIVITIES /KAT=GK /UE_PARAMS="<configured variant>"

Parameter name	**Value**
Comment:	Indirect activity allocation HYDRA → SAP
Interval	5
To upload the time tickets for indirect activity allocation - upload client	
Product key	SAP-COILV
License key	SAP-COILV
Command (Windows):	sh.exe ./hysapupl.scr /UPLSEGNAM=/UPLSEGNAM=E2ACC_SENDER_ACTIVITIES /SINGLE_IDOC /SUBLEVEL=2
Command (Unix):	./hysapupl.scr /UPLSEGNAM=/UPLSEGNAM=E2ACC_ACT_ALLOC000 /SINGLE_IDOC /SUBLEVEL=2
Comment:	Direct activity allocation HYDRA → SAP
Interval	5

10 Application Relevant Settings in SAP

Maintenance of the SAP partner agreement – inbound processing

Maintain the following settings for inbound processing in the partner agreement in SAP (WE20)

Parameter name	**Value**
Set-up for the direct activity allocation	
Partner number	Created logical system
Partner type	LS
Message type	ACC_ACT_ALLOC
Transaction code	BAPI
Appliance for direct activity allocation	
Partner number	Created logical system
Partner type	LS
Message type	ACC_SENDER_ACTIVITIES
Transaction code	BAPI

Maintenance of the SAP distribution model - inbound processing

Parameter name	**Value**
Set-up for the direct activity allocation	
Model view	Created model view
Sender/ client	Logical system for the sender system
Recipient/ server	Logical system of the client
Object name/ interface	AcctngActivityAlloc

Parameter name	**Value**
Method	Post
For indirect activity allocation	
Model view	Created model view
Sender/ client	Logical system for the sender system
Recipient/ server	Logical system of the client
Object name/ interface	AcctngSenderActivity
Method	Post

11 MYERPRCK - Program Parameters

Purpose

Use the upload program myerprck.exe/out to create confirmations/uploads to higher-level systems. In addition to the settings you make directly in the applications, you can also use program parameters to control confirmations/uploads.

Integration

The confirmation/upload is integrated with numerous components, for example:

- Shop floor data collection
- Tracking and tracing as well as material and production logistics
- Detailed scheduling

Available program parameters:

Parameters	Meaning/use	Relevant interfaces	Productive release
Program parameters to control processing:			
/MESTYP=XXXX	The parameter MESTYP defines the structure to be generated.	All	Yes
/GRP=XXXX	The grouping type specifies the criterion by which uploads should be grouped. Possible values: PLANT --> Groups by plant	Requires customizations	Requires customizations
/V=sssss	Since SAP R/3 PP does not support correction postings, HYDRA allows to retain confirmations/uploads for correction purposes in HYDRA for a specific period of time. Use the parameter /V=sssss (sssss =	EIS-ERP EIS-XPPS SAP-PPPDC SAP-PPREM	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	delay time in seconds) to activate the above described delay when the upload program is called. Examples: `myerprck.exe/out /V=3600` The system only uploads postings that are older than one hour.	SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	
/BIS=DDMMYYHHMM **/BIS=**HHMM /TILLDATE=MM/DD/YYYY /TILLTIME=sec after midnight	Use the parameter /BIS= DDMMYYHHMM (date + time) when calling the upload program to enter the delay as a point in time. You can enter this point in time with date and time or you can just enter the time in the format "HHMM". In the latter case, the time refers to the current day. `Myerprck.exe /BIS=2505110600` This parameter uploads postings that were recorded until 06:00 a.m. on 25 May 2011. `Myerprck.exe /BIS=0600` This parameter uploads postings that were recorded until 06:00 a.m. of the current day.	EIS-ERP EIS-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TZ=+/-sssss	Use the parameter /TZ=+/-sssss to adapt uploads to different time zones. The parameter adjusts the time specifications entered in the fields EXEC__START_TIME, EXEC_FIN_TIME and LOGTIME of the upload structure of the SAP-PPPDC interface according to its specifications.	SAP-PPPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KST=XXX	Use this parameter to restrict the data to be uploaded. In this case, the system only uploads data of a specified cost center. Use the parameter /KST=XXX (XXX = cost center, a max. of 8 characters) when calling the upload program myerprck.exe/out to enable the above-described restriction. Then the system only uploads data records that were posted to machines of the specified cost center. The system checks the cost center of the machine/workplace that is entered as the posting workplace/machine in the posting record. The system only checks the cost center of the workplace/machine. You can specify the parameter several times per call. <u>Example:</u> `Myerprck.exe /KST=BDE100 /KST=BDE200` The system only uploads records that were posted onto machines of the cost center BDE100/BDE200.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/CLEAR_RES	Use the parameter "/CLEAR_RES“ to assign an "X" to the field CLEAR_RES of the upload structure when it comes to a final confirmation/upload (record type L40). Consequently, SAP will clear open reservations for the respective order.	SAP-PPPDC	Yes
/NEG_MENGE	By default, quantities (L20/L40) cannot	SAP-PPPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	be uploaded to SAP PP using partial confirmations/uploads via the SAP-PPPDC interface if data is collected at the same time via the total quantity counter of MDE machines, since SAP is not able to process negative quantities. This type of collection can result in negative quantity postings for yield when OPs are finished. This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations). In this case, you can use the program parameter /NEG_MENGE to enable the upload of these quantities.		
/LA_MNR	The SAP_PMCC3 interface requires the activity type to be uploaded to SAP PM. The activity type can be identified via the machine/workplace where the posting was performed. Use this program parameter to enable identification of the activity type. Then the system uses the machine to identify the activity type from the activity types kept in HYDRA.	SAP-PMCC3	Yes
/IDENT_PRAEFIX=	In the upload structure of the SAP-PPPDC interface, the field EX_IDENT uniquely identifies uploads from subsystems. HYDRA populates the field. You can add a prefix to the EX_IDENT field to differentiate between uploads	SAP-PPPDC SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	from various HYDRA subsystems connected to one SAP instance. Example: `Myerprck.exe /IDENT_PRAEFIX=ABC` The prefix may only include hexadecimal characters: A –H und 0 – 9.		
/ABZEICH=XX	While customizing the order type, you can specify that only signed data records are uploaded. Use the parameter /ABZEICH=XX to specify a period of time in days after that you can upload even unsigned data records.	EIS-ERP EIS-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TRANSFER=	Use the parameter "/TRANSFER=" to only upload records whose specifications were transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/NOTRANSFER=XXX	Use the parameter "/NOTRANSFER=" to only upload records whose specifications were NOT transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SEK	The EIS-ERP interface uploads the times of resource performance accounts in hours. In particular with very short lead times this may effect that logon times are cut off by a conversion into hours. Use this program parameter to upload times in seconds.	EIS-ERP ESI-XPPS	Yes
/RMTYP=	When customizing the order type, you can assign an upload type to the order type. Use this program parameter to only upload data records of this upload type. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KAT=	When customizing the order type, you can connect the order type with a category. Use the program parameter /KAT= to only upload data records of this category. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SART=	The system only uploads ADE log postings of the specified record type. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the corresponding uploads when customizing the order type. You can specify the parameter several times per call. Example: `Myerprck.exe /SART=A /SART=E` ⇨ The system only uploads A and E records.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/NOLOCK	When starting the upload program, the system checks if there are any lock entries for the database table ADE_PROTOKOLL. If this is the case,	All	Requires customizations

Parameters	Meaning/use	Relevant interfaces	Productive release
	the upload is not carried out. You can use this program parameter to prevent this check. Set this parameter, in particular, if the upload is not based on the table ade_protokoll.		
/EINH_CC34	The interfaces SAP-PMCC3 and SAP-PSCC4 transfer the uploaded activity quantity in seconds (SEC) to SAP. Use the parameter **"/EINH_CC34“** to upload the data in other units. The following units are supported: Hours: H, HUR, STD Minutes: MIN Seconds: SEC Example: `Myerprck.exe /EINH_CC34=HUR` The system uploads the recorded times in the unit "HUR“ (hours).	SAP-PMCC3 SAP-PSCC4	Yes
/SDAT_STORNO	The SAP-PPPDCC interface transfers the change date along with the correction records. Use this program parameter to upload the initially collected shift date instead.	SAP-PPPDCC	Yes
/NORFC_STORNO	The SAP-PPPDCC interface transfers the cancellation records via sRFC. Use the program parameter to transfer the data in the IDoc format to SAP. To do	SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	so, inbound processing must be implemented in SAP. The system uploads the cancellation records via the standard PP-PDC segment (with record type K20/K40) as if the PP-PDCC license was not available.		
/PI	If you use the SAP Process Integration (previously: Exchange Infrastructure) to communicate with SAP, the version of the transferred segment is checked more strictly. Use the program parameter to transfer segment names with the version number (i.e. the trailing zeros of the segment name).	SAP-PPPDC SAP-PMCC3 SAP-PSCC4	Yes
/INDEX_TMP_TABLE	Use this parameter to accelerate uploads if ORACLE is used as database system and large amounts of data are affected. To do so, use an index for a temporary table where all data to be uploaded is transferred in a first step.	All	Requires customizations
/UE_PARAMS=	Program parameter for the stand-alone user exit processing (DD format).	Various	Yes
/NOSTORNO	Use this program parameter to prevent cancellation records from being uploaded. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the	All	Yes

Parameters	**Meaning/use**	**Relevant interfaces**	**Productive release**
	corresponding uploads when customizing the order type.		
/RECALC_NEG_YIELD	Use this parameter to offset negative yield with already posted positive uploads.	SAP-PPPDCC	Requires customizations
Program parameters to use the SIGUSR communication:			
/LOGGING	Use this program parameter to activate communication from the database table HYD_LOGGING. To do so, a customization might be required.	INDIVIDUAL CASE	Yes
/WAIT_SIGUSR1=XX	The program parameter specifies the time in seconds that has to pass before the upload is performed via the SIGUSR communication even without trigger.	INDIVIDUAL CASE	Yes
/PEEK_SIGUSR1=XX	Use this parameter to delay execution of an action triggered by the SIGUSR communication. The delay time is entered in seconds for this parameter. The program interprets this time as follows: If within the next second after the initial trigger there is another trigger, then wait for not more than <specified value> seconds. If in a specific case, triggers would indeed arrive every second then the	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	WAIT_SIGUSR time (e.g. 120 seconds) would apply; i.e. the system would in fact perform the upload after 2 minutes.		
/SEND_SIGUSR1=	This program parameter defines which other process/ program must be triggered after processing by the SIGUSR communication. Specify the process/program WITHOUT file extension.	INDIVIDUAL CASE	Yes
/COUNT_SIGUSR1=XX	Uploading in signal mode can hardly be subjected to tracing. This is due to the fact that the program in those cases is started once via the scheduler but won't shut off. Any redirection of the program call with -d to a log file will then necessarily lead to very large log files, which will negatively affect the performance. Use the new program parameter /COUNT_SIGUSR1=XX to specify after how many calls the program will automatically shut down. A call in these instances is both, a call via SIGUSR communication and the cyclical program execution which is controlled via the parameter /WAIT_SIGUSR1. Then the scheduler restarts the program. But this will lead to a time period "t" during which SIGUSR calls will not be processed. It is, however, assumed that this will not lead to data losses since the data to be uploaded are already saved to	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	the DB. Benefits: If the program is started via a script (*.scr) from the scheduler, you can store there the routine to generate a date/ time stamp file name for the log file to be created. This allows to restrict the log file size.		
Program parameters for debugging/ tracing/ testing/ logging purposes:			
/ONLYERR	This program parameter specifies that system log entries are only created if an error occurred during uploading. This reduces the entries in the system log.	All	Yes
/SIM	The system does not upload/confirm data during simulations (the uploaded/confirmed indicator is set to '"True").	All	No
/SIMULATION	The system does not upload/confirm data to SAP during simulation (confirmed/uploaded indicator will not be changed).	All	No